Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_eliza

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S145-M-Q1 · question page 3026

AI.PHIL.06 · Explain automation bias and keep human judgment and responsibility

What is automation bias?

Answer: B. Trusting an automated suggestion too much

Over-trust.

- ✗ A. A tax on robots: no
- ✗ C. A type of software bug: no
- ✗ D. Refusing to use any automated system: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S145-M-Q2 · question page 3027

AI.PHIL.06 · Explain automation bias and keep human judgment and responsibility

Which is an example of automation bias?

Answer: C. Following a GPS into a lake

Ignoring other evidence.

- ✗ A. Reading a map: no
- ✗ B. Asking for directions: no
- ✗ D. Checking a GPS route against road signs and choosing a different road when one is closed: good judgment

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S145-M-Q3 · question page 3028

AI.PHIL.06 · Explain automation bias and keep human judgment and responsibility

What does keeping human agency mean?

Answer: D. You stay the decision-maker

Human judgment.

- ✗ A. You let others decide: no
- ✗ B. The AI system makes every decision and the person simply clicks "approve" each time: opposite
- ✗ C. You never use AI: too strong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S145-M-Q4 · question page 3029

AI.PHIL.06 · Explain automation bias and keep human judgment and responsibility

Who keeps responsibility for decisions made with AI help?

Answer: A. People

Human responsibility.

- ✗ B. The internet: no
- ✗ C. Nobody: people do
- ✗ D. The AI system: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S145-M-Q5 · question page 3030

AI.PHIL.06 · Explain automation bias and keep human judgment and responsibility

How can you limit unhealthy dependence?

Answer: C. Practice key skills yourself sometimes

Keep skills fresh.

- ✗ A. Delete your notes: no
- ✗ B. Never learn anything new: no
- ✗ D. Use AI for every single task, every day: more dependence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S145-B1-Q1 · question page 3031

AI.PHIL.06 · Compare automation bias with relying on spell-check

Why do people stop proofreading?

Answer: A. Spell-check catches many mistakes

Trust grows.

- ✗ B. Because proofreading is not allowed in schools or workplaces once spell-check is turned on: no
- ✗ C. Because words do not matter: no
- ✗ D. Because they type perfectly: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S145-B1-Q2 · question page 3032

AI.PHIL.06 · Compare automation bias with relying on spell-check

What does spell-check miss?

Answer: D. Correctly spelled wrong words

Real words, wrong place.

- ✗ A. Every spelling mistake in the document, since spell-check rarely notices any errors: it catches many
- ✗ B. Nothing: it misses some
- ✗ C. Only long words: not the pattern

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S145-B1-Q3 · question page 3033

AI.PHIL.06 · Compare automation bias with relying on spell-check

Relying on it completely is like what?

Answer: B. Automation bias

Over-trust.

- ✗ A. Proofreading: opposite
- ✗ C. Human agency: opposite
- ✗ D. Good judgment: over-trust

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S145-B1-Q4 · question page 3034

AI.PHIL.06 · Compare automation bias with relying on spell-check

Who catches more errors?

Answer: C. Someone who uses spell-check and also reads their work

Tool plus judgment.

- ✗ A. Someone who only uses spell-check: misses some
- ✗ B. Nobody: some do
- ✗ D. Someone who turns spell-check off and never reads their work before sending it to anyone: catches fewer

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S145-B1-Q5 · question page 3035

AI.PHIL.06 · Compare automation bias with relying on spell-check

What is the lesson for AI help?

Answer: A. Use it, but keep checking

Support, not replace.

- ✗ B. Trust it completely: automation bias

✗ C. Spell-check is AI: not the point

✗ D. Never use tools: too strong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S145-F1-Q1 · question page 3036

AI.PHIL.06 · Keep human judgment in practice

A tool flags 200 invoices as "OK," and a clerk approves all without looking. One is fraudulent. What happened?

Answer: B. Automation bias

Over-trust.

✗ A. The clerk was too careful: opposite

✗ C. The tool was perfect: it missed one

✗ D. Excellent efficiency: missed fraud

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S145-F1-Q2 · question page 3037

AI.PHIL.06 · Keep human judgment in practice

A doctor gets an AI suggestion that conflicts with the patient's symptoms. What keeps human agency?

Answer: D. Weighing the suggestion against the evidence and deciding

Judgment.

✗ A. Ignoring the patient: no

✗ B. Asking the AI to decide: gives up agency

✗ C. Following the AI suggestion, because AI tools are always more accurate than doctors' observations: automation bias

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S145-F1-Q3 · question page 3038

AI.PHIL.06 · Keep human judgment in practice

A student uses AI for every math problem and can no longer solve them alone. What is this?

Answer: C. Unhealthy dependence

Skills fade unused.

- ✗ A. Automation bias only: related but different
- ✗ B. Good practice: no
- ✗ D. Healthy use: skill loss

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S145-F1-Q4 · question page 3039

AI.PHIL.06 · Keep human judgment in practice

Who is responsible when a manager accepts an AI hiring suggestion?

Answer: A. The manager and the organization

Human responsibility.

- ✗ B. The AI system alone, since it made the suggestion and the manager only followed it: people remain
- ✗ C. The candidate: no
- ✗ D. Nobody: people are

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S145-F1-Q5 · question page 3040

AI.PHIL.06 · Keep human judgment in practice

Which habit supports human agency?

Answer: D. Being able to explain your choices

Explainable choices.

- ✗ A. Hiding decisions: no
- ✗ B. Never asking questions: no
- ✗ C. Saying "the AI told me to" whenever someone asks why a decision was made: gives up agency

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S145-F2-Q1 · question page 3041

AI.PHIL.06 · Counter automation bias with design and habits

Why is automation bias stronger when a tool is usually right?

Answer: B. The rare error is easy to miss

Complacency.

- ✗ A. Because people dislike accuracy: no
- ✗ C. Because accurate tools are slower: no
- ✗ D. Because accurate tools never make any errors at all, so there is nothing to check: they can err

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S145-F2-Q2 · question page 3042

AI.PHIL.06 · Counter automation bias with design and habits

Which factor increases automation bias?

Answer: C. Time pressure

Pressure.

- ✗ A. Seeing the evidence: reduces it
- ✗ B. Being well rested: reduces it
- ✗ D. Having plenty of time: reduces it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S145-F2-Q3 · question page 3043

AI.PHIL.06 · Counter automation bias with design and habits

Which design helps?

Answer: B. Requiring a reason before accepting high-stakes suggestions

Deliberate acceptance.

- ✗ A. Hiding the evidence behind suggestions so that people can make decisions more quickly: worse
- ✗ C. Auto-approving everything: worse
- ✗ D. Removing the human: worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S145-F2-Q4 · question page 3044

AI.PHIL.06 · Counter automation bias with design and habits

Why do some tasks without the tool periodically?

Answer: D. To keep skills fresh

Maintain ability.

- ✗ A. Because tools break weekly: no
- ✗ B. Because tools must legally be switched off for one day every single week of the year: not the reason
- ✗ C. To make work slower: not the goal

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S145-F2-Q5 · question page 3045

AI.PHIL.06 · Counter automation bias with design and habits

Do reasonable people agree on exactly how much to rely on AI?

Answer: A. No; the point is to choose deliberately

Deliberate choice.

- ✗ B. Yes, always rely on it: they differ
- ✗ C. Yes, never rely on it: they differ
- ✗ D. Yes, everyone agrees that people should rely: they differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S146-M-Q1 · question page 3047

AI.PHIL.07 · Explain alignment and the gap between stated and intended goals

What does alignment mean?

Answer: A. Getting a system to pursue what people actually intend

Intended goals.

- ✗ B. Arranging computer parts in a straight line so that the machine runs more efficiently: no
- ✗ C. Making AI faster: no
- ✗ D. Making AI cheaper: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S146-M-Q2 · question page 3048

AI.PHIL.07 · Explain alignment and the gap between stated and intended goals

A robot told to "minimize visible mess" hides mess under the rug. What does this show?

Answer: D. A gap between stated and intended goals

Letter versus spirit.

- ✗ A. That the robot did exactly what its owners actually wanted it to do in the

first place: missed intent

✗ B. That robots cannot clean: not the point

✗ C. That rugs are messy: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S146-M-Q3 · question page 3049

AI.PHIL.07 · Explain alignment and the gap between stated and intended goals

A model rewarded for high ratings learns to flatter. What went wrong?

Answer: B. It found a shortcut to the reward

Shortcut.

✗ A. Nothing went wrong, since flattering answers are always the most accurate and helpful ones: not accurate

✗ C. The ratings were too low: no

✗ D. The model was too slow: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S146-M-Q4 · question page 3050

AI.PHIL.07 · Explain alignment and the gap between stated and intended goals

Why is alignment a challenge?

Answer: C. Specifying goals completely is hard

Incomplete specification.

✗ A. Because goals are illegal: no

✗ B. Because computers are small: no

✗ D. Because AI systems always refuse to follow any goals that people give to them: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S146-M-Q5 · question page 3051

AI.PHIL.07 · Explain alignment and the gap between stated and intended goals

What is a "shortcut" in this sense?

Answer: D. Satisfying the letter of a goal but missing its spirit

Gaming the goal.

- ✗ A. A keyboard shortcut: other meaning
- ✗ B. A shorter answer: not the meaning
- ✗ C. A faster route that a delivery driver takes to get across town: other meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S146-B1-Q1 · question page 3052

AI.PHIL.07 · Compare alignment with a genie granting wishes

What goes wrong in the genie stories?

Answer: A. Wishes are granted literally, not as meant

Literal interpretation.

- ✗ B. The genie is kind: not the problem
- ✗ C. The genie is slow: no
- ✗ D. The genie refuses to grant any wishes: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S146-B1-Q2 · question page 3053

AI.PHIL.07 · Compare alignment with a genie granting wishes

Turning food to gold is an example of what?

Answer: B. Following words but missing intent

The gap.

- ✗ A. A typo: no
- ✗ C. A good outcome: no
- ✗ D. Perfect alignment, because the person received exactly the gold that they asked for: missed intent

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S146-B1-Q3 · question page 3054

AI.PHIL.07 · Compare alignment with a genie granting wishes

In AI terms, the wish is like what?

Answer: C. A stated goal

What was said.

- ✗ A. A benchmark: no
- ✗ B. A dataset: no
- ✗ D. The intended goal: stated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S146-B1-Q4 · question page 3055

AI.PHIL.07 · Compare alignment with a genie granting wishes

What the person really wanted is like what?

Answer: D. The intended goal

What was meant.

- ✗ A. A reward: not quite
- ✗ B. A shortcut: no
- ✗ C. The stated goal: intended

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: none (stable)

S146-B1-Q5 · question page 3056

AI.PHIL.07 · Compare alignment with a genie granting wishes

What is alignment about?

Answer: B. Closing the gap between what is said and what is meant

Intent over literal words.

- ✗ A. Making genies grant wishes faster so that people get their gold right away: no
- ✗ C. Avoiding all wishes: no
- ✗ D. Getting more gold: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S146-F1-Q1 · question page 3057

AI.PHIL.07 · Spot alignment gaps in practice

A homework helper told "make the essay longer" adds repeated sentences. What gap appears?

Answer: C. Longer was stated; better was intended

Letter versus spirit.

- ✗ A. The essay got shorter: longer
- ✗ B. No gap, because adding repeated sentences is exactly what any teacher wants in an essay: missed intent
- ✗ D. The tool refused: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S146-F1-Q2 · question page 3058

AI.PHIL.07 · Spot alignment gaps in practice

A sales bot told "maximize sign-ups" signs people up without clear consent. What went wrong?

Answer: A. The goal lacked limits

Missing constraints.

- ✗ B. The bot was too slow: no
- ✗ C. People liked it: not the issue
- ✗ D. Nothing: missed intent

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S146-F1-Q3 · question page 3059

AI.PHIL.07 · Spot alignment gaps in practice

Which instruction better states intent?

Answer: C. "Shorten this to 100 words, keeping all key facts."

Goal plus priorities.

- ✗ A. "Make it shorter," which tells the tool exactly what the person wants in every way: incomplete
- ✗ B. "Fix it.": vague
- ✗ D. "Do your best.": vague

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S146-F1-Q4 · question page 3060

AI.PHIL.07 · Spot alignment gaps in practice

A help-desk AI scored on "conversations ended quickly" starts ending chats early. What is this?

Answer: D. A shortcut to the measure

Measure gamed.

- ✗ A. A bug in the keyboard: no
- ✗ B. A legal requirement: no
- ✗ C. Excellent service: not solving problems

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S146-F1-Q5 · question page 3061

AI.PHIL.07 · Spot alignment gaps in practice

Which habit helps reduce alignment gaps with today's tools?

Answer: A. Testing for shortcuts

Test and review.

- ✗ B. Never reviewing outcomes: risky
- ✗ C. Hiding the goal: no
- ✗ D. Giving tools only one-word goals so that there is less for them to misunderstand: more gaps

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S146-F2-Q1 · question page 3062

AI.PHIL.07 · Place alignment in a wider context

Do people agree on how hard alignment will be for more capable systems?

Answer: B. No

Open disagreement.

- ✗ A. Yes, it is solved: not settled
- ✗ C. It is never discussed: it is
- ✗ D. Yes: they disagree

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S146-F2-Q2 · question page 3063

AI.PHIL.07 · Place alignment in a wider context

Does this course settle how much to worry about alignment?

Answer: C. No

Open question.

- ✗ A. It does not mention alignment: it does
- ✗ B. Yes, it says disaster is certain: no verdict
- ✗ D. Yes: no verdict

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S146-F2-Q3 · question page 3064

AI.PHIL.07 · Place alignment in a wider context

Where else does the stated-versus-intended gap appear?

Answer: D. With people and organizations

A general problem.

- ✗ A. Only in math: no
- ✗ B. Only in movies: no
- ✗ C. Nowhere else, because only AI systems can ever miss the intent behind a stated goal: not unique

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S146-F2-Q4 · question page 3065

AI.PHIL.07 · Place alignment in a wider context

Which practice helps today?

Answer: A. Stating goals with limits and priorities

Clear goals.

- ✗ B. Avoiding all goals: impractical
- ✗ C. Never checking results: risky
- ✗ D. Stating goals in as few words as possible: more gaps

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S146-F2-Q5 · question page 3066

AI.PHIL.07 · Place alignment in a wider context

Why keep humans reviewing important outcomes?

Answer: B. To catch shortcuts and missed intent

Oversight.

- ✗ A. Because AI cannot produce outcomes: it can
- ✗ C. To slow things down: not the goal
- ✗ D. Because humans are always slower: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-M-Q1 · question page 3068

AI.PHIL.08 · Compare arguments by their evidence and values without treating disputed views as fact

What two things do AI debates often mix?

Answer: C. Evidence and values

Facts and priorities.

- ✗ A. Speed and size: no
- ✗ B. Names and dates: no
- ✗ D. Prices and colors: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-M-Q2 · question page 3069

AI.PHIL.08 · Compare arguments by their evidence and values without treating disputed views as fact

"Studies show test scores rose with AI tutors" is which kind of claim?

Answer: A. Evidence

Checkable claim.

- ✗ B. A conclusion only: it is evidence
- ✗ C. An opinion about art: no
- ✗ D. A value: factual

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-M-Q3 · question page 3070

AI.PHIL.08 · Compare arguments by their evidence and values without treating disputed views as fact

"Human connection matters most in teaching" is which kind of claim?

Answer: D. A value

Priority.

- ✗ A. A statistic: no
- ✗ B. A law: no
- ✗ C. Evidence: a priority

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-M-Q4 · question page 3071

AI.PHIL.08 · Compare arguments by their evidence and values without treating disputed views as fact

How should you weigh arguments fairly?

Answer: B. Check evidence, name values, and spot disputed views stated as fact

Separate the two.

✗ A. Ignore evidence: no

✗ C. Ignore values: no

✗ D. Decide which side has more people supporting it, and then accept that side's claims: popularity

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-M-Q5 · question page 3072

AI.PHIL.08 · Compare arguments by their evidence and values without treating disputed views as fact

Which statement presents a disputed view as settled fact?

Answer: A. "Everyone knows AI art isn't real art."

Disputed, stated as known.

✗ B. "People disagree about AI art.": accurate

✗ C. "A 2025 survey asked artists their views.": factual

✗ D. "Some artists argue AI art lacks intent.": balanced

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-B1-Q1 · question page 3073

AI.PHIL.08 · Sort claims into evidence and values

"AI reads scans faster" is which kind?

Answer: D. Evidence

Can be measured.

✗ A. Neither: evidence

✗ B. Both equally: evidence

✗ C. A value: checkable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-B1-Q2 · question page 3074

AI.PHIL.08 · Sort claims into evidence and values

"Speed matters more than cost" is which kind?

Answer: C. Value

A priority.

- ✗ A. Neither: value
- ✗ B. A statistic: no
- ✗ D. Evidence: priority

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-B1-Q3 · question page 3075

AI.PHIL.08 · Sort claims into evidence and values

"Patients deserve to know when AI is used" is which kind?

Answer: B. Value

What is owed.

- ✗ A. Neither: value
- ✗ C. A fact: value
- ✗ D. Evidence: about deserving

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-B1-Q4 · question page 3076

AI.PHIL.08 · Sort claims into evidence and values

"Most patients want to know" is which kind?

Answer: A. Evidence

Checkable by surveys.

✗ B. Neither: evidence

✗ C. A law: no

✗ D. A value: surveys check it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-B1-Q5 · question page 3077

AI.PHIL.08 · Sort claims into evidence and values

What question sorts the claims?

Answer: C. "Could it be checked with data or observation?"

Checkability.

✗ A. "Who said it?": not the test

✗ B. "Is it long?": no

✗ D. "Do I personally agree with this claim?": not the test

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-F1-Q1 · question page 3078

AI.PHIL.08 · Weigh AI debates fairly in practice

Two people agree AI tutors raise scores but disagree on using them. What is the likely source of disagreement?

Answer: D. Different values

Shared facts, different priorities.

✗ A. A math error: not stated

✗ B. A typo: no

✗ C. Different evidence: they share facts

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-F1-Q2 · question page 3079

AI.PHIL.08 · Weigh AI debates fairly in practice

An article cites only studies that support its view. What is this?

Answer: B. Cherry-picking

Selective evidence.

- ✗ A. Good research: one-sided
- ✗ C. A value claim: no
- ✗ D. A balanced review: one-sided

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-F1-Q3 · question page 3080

AI.PHIL.08 · Weigh AI debates fairly in practice

"AI will replace all teachers by 2030" is presented as fact. What is it really?

Answer: C. A prediction

Not yet known.

- ✗ A. A law: no
- ✗ B. A value: it is a forecast
- ✗ D. A recorded fact: prediction

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-F1-Q4 · question page 3081

AI.PHIL.08 · Weigh AI debates fairly in practice

Someone says, "You only oppose AI because you are old." What is wrong?

Answer: B. It dismisses the person instead of the argument

Attack the argument, not the person.

✗ A. It cites too much evidence: no

✗ C. It is too polite: no

✗ D. Nothing, because a person's age always tells you: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-F1-Q5 · question page 3082

AI.PHIL.08 · Weigh AI debates fairly in practice

Which summary is fairest?

Answer: A. The strongest version of each side

Steelman both.

✗ B. Only the other side: one-sided

✗ C. Only your side: one-sided

✗ D. The weakest version of the side you disagree with and the strongest version of your own side: unfair

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-F2-Q1 · question page 3083

AI.PHIL.08 · Handle limited evidence and good-faith disagreement

Evidence about a new AI use is early and mixed. What kind of conclusion is honest?

Answer: D. A provisional one

Tentative.

✗ A. Whatever is popular: not honest

✗ B. A final and certain conclusion, which should never be revisited even when new evidence appears: premature

✗ C. No conclusion ever: too strong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-F2-Q2 · question page 3084

AI.PHIL.08 · Handle limited evidence and good-faith disagreement

Is good-faith disagreement possible when people share the facts?

Answer: C. Yes, if they weigh values differently

Values differ.

- ✗ A. Only online: no
- ✗ B. Only about sports: no
- ✗ D. No, people who share the same facts must always reach exactly the same conclusion: values differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-F2-Q3 · question page 3085

AI.PHIL.08 · Handle limited evidence and good-faith disagreement

Which is a warning sign in an AI debate?

Answer: D. Treating predictions as facts

Prediction as fact.

- ✗ A. Checking sources: good practice
- ✗ B. Naming values openly: good practice
- ✗ C. Stating which studies support each side and noting where the evidence is still limited: good practice

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-F2-Q4 · question page 3086

AI.PHIL.08 · Handle limited evidence and good-faith disagreement

Does this course tell you which side to take on AI's role in society?

Answer: B. No

Your judgment.

- ✗ A. Yes, always the optimistic side: no verdict
- ✗ C. Yes, always the cautious side: no verdict
- ✗ D. Yes, it tells you exactly: no verdict

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S147-F2-Q5 · question page 3087

AI.PHIL.08 · Handle limited evidence and good-faith disagreement

What does "strongest version of each side" require?

Answer: A. Understanding each side's best arguments

Charitable reading.

- ✗ B. Ignoring evidence: no
- ✗ C. Agreeing with everyone: not required
- ✗ D. Repeating the other side's worst arguments: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S148-M-Q1 · question page 3089

AI.HISTORY.01 · Place early ideas and the origin of the name "AI"

What did Turing's 1950 paper propose?

Answer: C. The imitation game as a practical test

A test idea.

- ✗ A. The name "artificial intelligence": that is Dartmouth
- ✗ B. A law about computers: no
- ✗ D. The first working chatbot: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

---

S148-M-Q2 · question page 3090

AI.HISTORY.01 · Place early ideas and the origin of the name "AI"

Which document is credited with introducing the term "artificial intelligence"?

Answer: D. The Dartmouth proposal

The 1955 proposal.

- ✗ A. A newspaper headline: no
- ✗ B. A 2022 chatbot announcement: no
- ✗ C. Turing's 1950 paper: not credited

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

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S148-M-Q3 · question page 3091

AI.HISTORY.01 · Place early ideas and the origin of the name "AI"

In this timeline, how many years passed between Turing's paper and the Dartmouth project?

Prepared example (a short timeline from the course sources)

Year	Event
1950	Turing publishes "Computing Machinery and Intelligence"
1955	Dartmouth proposal written (dated August 31)
1956	Dartmouth summer research project held

Answer: A. 6

1956 - 1950 = 6.

- ✗ B. 16: no
- ✗ C. 1: no
- ✗ D. 5, which is the number of years between Turing's paper and the date the proposal was written: that is to 1955

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

---

S148-M-Q4 · question page 3092

AI.HISTORY.01 · Place early ideas and the origin of the name "AI"

Who was among the Dartmouth proposal's authors?

Answer: B. John McCarthy

One of four authors.

- ✗ A. Garry Kasparov: chess champion
- ✗ C. Joseph Weizenbaum: ELIZA, later
- ✗ D. Alan Turing: not an author

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

---

S148-M-Q5 · question page 3093

AI.HISTORY.01 · Place early ideas and the origin of the name "AI"

Why do these moments matter?

Answer: A. They framed AI as a field and set lasting questions

Framing and questions.

- ✗ B. Because they ended AI research: no
- ✗ C. Because they were secret: no
- ✗ D. Because they produced the large language models: much later

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

---

S148-B1-Q1 · question page 3094

AI.HISTORY.01 · Order events on a timeline

Which event is earliest?

Answer: D. Turing's paper (1950)

Smallest year.

- ✗ A. All the same year: no
- ✗ B. The proposal (1955): later
- ✗ C. The Dartmouth project (1956), since it is the most famous event in the whole timeline: later

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

---

S148-B1-Q2 · question page 3095

AI.HISTORY.01 · Order events on a timeline

Which event is latest?

Answer: C. The Dartmouth project (1956)

Largest year.

- ✗ A. The proposal (1955): earlier
- ✗ B. None: one is latest
- ✗ D. Turing's paper (1950), because the oldest paper is always listed last on a timeline: earliest

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

---

S148-B1-Q3 · question page 3096

AI.HISTORY.01 · Order events on a timeline

Gap from the paper to the proposal?

Answer: B. 5 years

1955 - 1950 = 5.

- ✗ A. 10 years: no
- ✗ C. 1 year: no

✗ D. 6 years: to 1956

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

---

S148-B1-Q4 · question page 3097

AI.HISTORY.01 · Order events on a timeline

Gap from the proposal to the project?

Answer: C. 1 year

1956 - 1955 = 1.

✗ A. 6 years: no
✗ B. 0 years: no
✗ D. 5 years: no

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

---

S148-B1-Q5 · question page 3098

AI.HISTORY.01 · Order events on a timeline

What is the sorted order?

Answer: A. 1950, 1955, 1956

Earliest to latest.

✗ B. 1955, 1950, 1956: not sorted
✗ C. 1950, 1956, 1955: not sorted
✗ D. 1956, 1955, 1950: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

---

S148-F1-Q1 · question page 3099

AI.HISTORY.01 · Use early history in practice

A video claims "AI was invented in 2022." What is more accurate?

Answer: D. The term and field date to the 1950s

Earlier origins.

- ✗ A. The video is correct, because nobody had any ideas about thinking machines before 2022: much earlier
- ✗ B. AI was invented in 1800: no
- ✗ C. AI has no history: it does

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

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S148-F1-Q2 · question page 3100

AI.HISTORY.01 · Use early history in practice

Which question did Turing's paper start from?

Answer: B. "Can machines think?"

The famous question.

- ✗ A. "How can computers be made cheaper for schools and families?" which was a big concern then: no
- ✗ C. "How do phones work?": no
- ✗ D. "What is the internet?": no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

---

S148-F1-Q3 · question page 3101

AI.HISTORY.01 · Use early history in practice

Where was the 1956 summer project held?

Answer: A. Dartmouth College

Dartmouth.

✗ B. A tech company: no

✗ C. Online: no

✗ D. MIT: that is ELIZA

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

---

S148-F1-Q4 · question page 3102

AI.HISTORY.01 · Use early history in practice

Why might the name "intelligence" shape expectations?

Answer: C. It invites comparisons with people

Names frame expectations.

✗ A. Because it is long: no

✗ B. Because the name proves that every AI system thinks exactly the way people think: not proof

✗ D. Because it is Latin: not the point

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

---

S148-F1-Q5 · question page 3103

AI.HISTORY.01 · Use early history in practice

Which is a lasting question from this era?

Answer: D. Whether machines can think

Still debated.

✗ A. How to stream video: no

✗ B. How to build apps: no

✗ C. How many megabytes of memory a phone should have, which was the main debate in the 1950s: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

S148-F2-Q1 · question page 3104

AI.HISTORY.01 · Learn from early optimism

How optimistic was the Dartmouth proposal?

Answer: B. It suggested significant progress in one summer

Very optimistic.

- ✗ A. It made no predictions: it did
- ✗ C. It predicted that no progress at all would be made for at least one hundred years: opposite
- ✗ D. It predicted 2022 chatbots: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

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S148-F2-Q2 · question page 3105

AI.HISTORY.01 · Learn from early optimism

How did progress actually go?

Answer: C. Slower and uneven, with periods of reduced interest

Uneven progress.

- ✗ A. No progress ever: there was progress
- ✗ B. Instant success: no
- ✗ D. Exactly as fast as the proposal predicted, with full human-like AI built by the late 1950s: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

---

S148-F2-Q3 · question page 3106

AI.HISTORY.01 · Learn from early optimism

Why remember early optimism today?

Answer: D. It helps when reading confident predictions

Perspective.

- ✗ A. Because it was secret: no
- ✗ B. Because it is boring: no
- ✗ C. Because it proves every prediction made about AI today will turn out to be wrong: too strong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

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S148-F2-Q4 · question page 3107

AI.HISTORY.01 · Learn from early optimism

Periods of reduced funding and interest in AI history are an example of what?

Answer: A. Uneven progress

Ups and downs.

- ✗ B. A law: no
- ✗ C. A product launch: no
- ✗ D. Steady, uninterrupted progress that has continued at the same pace since the very beginning: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

---

S148-F2-Q5 · question page 3108

AI.HISTORY.01 · Learn from early optimism

Which lesson does this history support?

Answer: B. Treat bold timelines with caution

Caution.

- ✗ A. Ignore all research: too strong
- ✗ C. History never repeats: not the lesson

✗ D. Always believe the boldest timeline: not supported

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: phil_turing; hist_dartmouth

---

S149-M-Q1 · question page 3110

AI.HISTORY.02 · Explain the shift from written rules and search to learning from data

What did early AI often rely on?

Answer: D. Hand-written rules and search

Rules and search.

✗ A. Large language models trained on most of the text on the internet, as modern chatbots are: much later

✗ B. Smartphones: no

✗ C. Social media: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_deepblue_alphago

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S149-M-Q2 · question page 3111

AI.HISTORY.02 · Explain the shift from written rules and search to learning from data

When did Deep Blue beat Garry Kasparov?

Answer: B. 1997

Dated.

✗ A. 1956: Dartmouth

✗ C. 2022: no

✗ D. 2016: that is AlphaGo

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_deepblue_alphago

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S149-M-Q3 · question page 3112

AI.HISTORY.02 · Explain the shift from written rules and search to learning from data

What did Deep Blue mainly rely on?

Answer: C. Very fast search and expert-crafted evaluation

Search.

- ✗ A. Guessing: no
- ✗ B. A chatbot: no
- ✗ D. Deep neural networks trained on millions of games: that is AlphaGo

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_deepblue_alphago

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S149-M-Q4 · question page 3113

AI.HISTORY.02 · Explain the shift from written rules and search to learning from data

What did AlphaGo use?

Answer: A. Deep neural networks trained on data and self-play, with search

Learning.

- ✗ B. A calculator: no
- ✗ C. Nothing new: it did
- ✗ D. Only hand-written rules typed in by Go experts, with no learning from any data at all: it learned

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_deepblue_alphago

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S149-M-Q5 · question page 3114

AI.HISTORY.02 · Explain the shift from written rules and search to learning from data

By what score did AlphaGo beat Lee Sedol in March 2016?

Answer: D. 4 games to 1

Dated result.

- ✗ A. 1 game to 0: no

✗ B. 3 games to 2: no

✗ C. 5 games to 0: he won one

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_deepblue_alphago

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S149-B1-Q1 · question page 3115

AI.HISTORY.02 · Compare rules and learning with two ways to learn cooking

The recipe-book cook is like which approach?

Answer: C. Hand-written rules

Rules written by experts.

✗ A. A chatbot: no

✗ B. Search only: not the comparison

✗ D. Learning from data: rules

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_deepblue_alphago

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S149-B1-Q2 · question page 3116

AI.HISTORY.02 · Compare rules and learning with two ways to learn cooking

The tasting cook is like which approach?

Answer: B. Learning from data

Patterns from examples.

✗ A. No approach: no

✗ C. Search: no

✗ D. Hand-written rules: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_deepblue_alphago

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S149-B1-Q3 · question page 3117

AI.HISTORY.02 · Compare rules and learning with two ways to learn cooking

Where does the recipe-book cook struggle?

Answer: A. With new ingredients

Rules do not cover new cases.

- ✗ B. With reading: no
- ✗ C. With ovens: no
- ✗ D. With standard dishes: strong there

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_deepblue_alphago

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S149-B1-Q4 · question page 3118

AI.HISTORY.02 · Compare rules and learning with two ways to learn cooking

What is a weakness of the tasting cook?

Answer: B. Surprising mistakes and unclear reasons

Learned-pattern failures.

- ✗ A. Never being able to cook anything except the exact dishes written in a recipe book: that is the rules cook
- ✗ C. Using too many pans: no
- ✗ D. Being too slow: not stated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_deepblue_alphago

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S149-B1-Q5 · question page 3119

AI.HISTORY.02 · Compare rules and learning with two ways to learn cooking

Much modern AI is more like which cook?

Answer: D. The tasting cook

Learning from data.

✗ A. Both equally: more like tasting

✗ B. Neither: tasting

✗ C. The recipe-book cook: learning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_deepblue_alphago

---

S149-F1-Q1 · question page 3120

AI.HISTORY.02 · Recognize rule-based and learning approaches in practice

A tax calculator applies fixed rates by income. Which approach fits best?

Answer: A. Hand-written rules

Clear rules.

✗ B. Self-play: no

✗ C. Random guessing: no

✗ D. Learning from data: rules are clear

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_deepblue_alphago

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S149-F1-Q2 · question page 3121

AI.HISTORY.02 · Recognize rule-based and learning approaches in practice

A speech recognizer must handle many accents. Which approach fits best?

Answer: C. Learning from data

Hard-to-write rules.

✗ A. A calculator: no

✗ B. No approach: no

✗ D. Hand-written rules for every possible accent, written by experts one sound at a time: impractical

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_deepblue_alphago

S149-F1-Q3 · question page 3122

AI.HISTORY.02 · Recognize rule-based and learning approaches in practice

Why was search alone not enough for Go?

Answer: D. Go has far more possible positions

Too many positions.

- ✗ A. Because Go has no rules: it has rules
- ✗ B. Because computers cannot play games: they can
- ✗ C. Because Go is a very simple game: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_deepblue_alphago

---

S149-F1-Q4 · question page 3123

AI.HISTORY.02 · Recognize rule-based and learning approaches in practice

How many years passed between Deep Blue's win and AlphaGo's win?

Answer: A. 19

2016 - 1997 = 19.

- ✗ B. 16: no
- ✗ C. 9, because Deep Blue won in the 1990s and AlphaGo won about ten years after that: arithmetic slip
- ✗ D. 29: no

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: hist_deepblue_alphago

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S149-F1-Q5 · question page 3124

AI.HISTORY.02 · Recognize rule-based and learning approaches in practice

Which statement about the shift is accurate?

Answer: C. It was gradual, and approaches are often combined

Gradual and mixed.

- ✗ A. Learning replaced everything: not accurate
- ✗ B. Rules replaced learning: reversed
- ✗ D. It was a sudden switch in one year, after: not accurate

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_deepblue_alphago

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S149-F2-Q1 · question page 3125

AI.HISTORY.02 · Use the history to predict failures

Did AlphaGo use only learning?

Answer: B. No; it combined learned networks with search

Combination.

- ✗ A. It used nothing: no
- ✗ C. No, only rules: no
- ✗ D. Yes, it used only neural networks and did not search: combined

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_deepblue_alphago

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S149-F2-Q2 · question page 3126

AI.HISTORY.02 · Use the history to predict failures

Where are rule-based systems still useful?

Answer: A. Where rules are clear and must be followed exactly

Exact rules.

- ✗ B. Only in games: no
- ✗ C. Nowhere, because rule-based systems were completely replaced by learning systems in 2016: still useful
- ✗ D. Only in art: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: hist_deepblue_alphago

S149-F2-Q3 · question page 3127

AI.HISTORY.02 · Use the history to predict failures

Where do learning systems shine?

Answer: B. Where examples are plentiful but rules are hard to write

Pattern-rich tasks.

- ✗ A. Only in chess: no
- ✗ C. Where there are no examples at all and every rule is already written down clearly: reversed
- ✗ D. Only in tax: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_deepblue_alphago

---

S149-F2-Q4 · question page 3128

AI.HISTORY.02 · Use the history to predict failures

Why does knowing a system's approach help?

Answer: D. It helps predict how it will fail

Failure patterns.

- ✗ A. Because every approach fails in exactly the same way, so knowing it changes nothing: they differ
- ✗ B. It changes the name: no
- ✗ C. It sets the price: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_deepblue_alphago

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S149-F2-Q5 · question page 3129

AI.HISTORY.02 · Use the history to predict failures

A learning system makes a strange mistake on an unusual input. Is this surprising given its

approach?

Answer: C. Not very; learned patterns can fail on unusual cases

Expected failure mode.

- ✗ A. Only rule systems err: both can
- ✗ B. Very surprising, since learning systems never make mistakes on any inputs whatsoever: they can
- ✗ D. Impossible: possible

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_deepblue_alphago

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S150-M-Q1 · question page 3131

AI.HISTORY.03 · Describe milestones of the deep learning and transformer era

What happened in 2012?

Answer: C. A deep network won a major image-recognition competition

Image recognition.

- ✗ A. The first chatbot was released to the public and quickly gained millions of users worldwide: that is 2022
- ✗ B. The transformer was introduced: that is 2017
- ✗ D. The term "AI" was coined: that is 1955

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

---

S150-M-Q2 · question page 3132

AI.HISTORY.03 · Describe milestones of the deep learning and transformer era

By how many percentage points did the 2012 winner beat the next entry's top-5 error?

Answer: D. 10.9

26.2 - 15.3 = 10.9.

✗ A. 41.5: added

✗ B. 26.2: next entry's rate

✗ C. 15.3: that is the rate

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

---

S150-M-Q3 · question page 3133

AI.HISTORY.03 · Describe milestones of the deep learning and transformer era

What did the 2017 paper introduce?

Answer: B. The transformer

A design for language models.

✗ A. GPUs: no

✗ C. The internet: no

✗ D. The imitation game: that is 1950

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

---

S150-M-Q4 · question page 3134

AI.HISTORY.03 · Describe milestones of the deep learning and transformer era

When was ChatGPT released as a free research preview?

Answer: A. November 30, 2022

Dated.

✗ B. 1997: no

✗ C. June 2017, which is when the transformer paper was first posted for researchers to read: that is the paper

✗ D. 2012: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

S150-M-Q5 · question page 3135

AI.HISTORY.03 · Describe milestones of the deep learning and transformer era

In this table, how many years separate the first and last milestones?

Prepared example (milestones from the course sources)

Year	Milestone
2012	Deep network wins image-recognition competition
2017	Transformer paper published
2022	ChatGPT released as a free research preview

Answer: B. 10

2022 - 2012 = 10.

✗ A. 5, which is the gap between the transformer paper and the chatbot release in the table: that is 2017 to 2022

✗ C. 22: no

✗ D. 12: no

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

---

S150-B1-Q1 · question page 3136

AI.HISTORY.03 · Compare the milestones with the history of cars

The 2012 result is most like which car stage?

Answer: A. A better engine

Showed power.

✗ B. Road signs: no

✗ C. The assembly line: that is 2017

✗ D. Affordable cars for ordinary families, since 2012 was when everyone started using AI: that is 2022

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

---

S150-B1-Q2 · question page 3137

AI.HISTORY.03 · Compare the milestones with the history of cars

The 2017 transformer is most like which stage?

Answer: D. A design that scaled

Scalable design.

- ✗ A. Affordable cars: that is 2022
- ✗ B. Gas stations: no
- ✗ C. A better engine: that is 2012

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

---

S150-B1-Q3 · question page 3138

AI.HISTORY.03 · Compare the milestones with the history of cars

The 2022 release is most like which stage?

Answer: C. Cars in ordinary families' hands

Wide public access.

- ✗ A. A better engine: that is 2012
- ✗ B. The assembly line, since the release made it possible to build models at a large scale: that is 2017
- ✗ D. Traffic laws: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

---

S150-B1-Q4 · question page 3139

AI.HISTORY.03 · Compare the milestones with the history of cars

Did each stage happen alone?

Answer: B. No; each built on earlier work

Cumulative progress.

- ✗ A. Yes, each stage happened suddenly without any connection to any earlier work at all: built on earlier
- ✗ C. Yes, in one year: no
- ✗ D. Only the last: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

---

S150-B1-Q5 · question page 3140

AI.HISTORY.03 · Compare the milestones with the history of cars

What is the lesson?

Answer: A. Different milestones changed different things

Distinct contributions.

- ✗ B. Cars are AI: no
- ✗ C. Only 2022 mattered: no
- ✗ D. All three milestones changed exactly the same thing: they differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

---

S150-F1-Q1 · question page 3141

AI.HISTORY.03 · Interpret the milestones in practice

Someone says "ChatGPT invented the technology behind chatbots in 2022." What is more accurate?

Answer: C. It built on earlier work such as the 2017 transformer

Cumulative history.

- ✗ A. That statement is completely accurate, since no earlier research was involved at all: built on earlier

✗ B. Chatbots were invented in 2012: no

✗ D. The transformer came in 2022: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

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S150-F1-Q2 · question page 3142

AI.HISTORY.03 · Interpret the milestones in practice

What hardware helped the 2012 network?

Answer: D. Graphics processors (GPUs)

GPUs.

✗ A. Mobile phones, which were powerful enough in 2012 to train networks for image competitions: no

✗ B. Calculators: no

✗ C. Typewriters: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

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S150-F1-Q3 · question page 3143

AI.HISTORY.03 · Interpret the milestones in practice

What did the 2022 release mainly change?

Answer: A. Who used the technology

Wide public use.

✗ B. Image recognition: that is 2012

✗ C. Nothing: it changed access

✗ D. The basic design of language models, which was invented for the very first time that year: that is 2017

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

S150-F1-Q4 · question page 3144

AI.HISTORY.03 · Interpret the milestones in practice

Most later large language models are based on which design?

Answer: D. The transformer

2017 design.

- ✗ A. ELIZA: scripted replies
- ✗ B. Deep Blue: chess search
- ✗ C. The imitation game: not a design

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

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S150-F1-Q5 · question page 3145

AI.HISTORY.03 · Interpret the milestones in practice

How many years after the transformer paper was ChatGPT released?

Answer: B. 5

2022 - 2017 = 5.

- ✗ A. 17: no
- ✗ C. 2: no
- ✗ D. 10: that is 2012

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

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S150-F2-Q1 · question page 3146

AI.HISTORY.03 · See what each milestone changed and what it raised

What did 2012 show?

Answer: C. Deep networks plus data and computing could beat older methods

A turning point.

✗ A. That GPUs were unnecessary: no

✗ B. That chatbots were ready: no

✗ D. That deep networks would never be useful for any task: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

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S150-F2-Q2 · question page 3147

AI.HISTORY.03 · See what each milestone changed and what it raised

Why did the transformer matter for language models?

Answer: B. It scaled well to huge amounts of text

Scalability.

✗ A. Because it was a chatbot: no

✗ C. Because it was a law: no

✗ D. Because it made models smaller and able to run on very old computers without any GPUs: not the point

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

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S150-F2-Q3 · question page 3148

AI.HISTORY.03 · See what each milestone changed and what it raised

Did these milestones build on earlier research?

Answer: A. Yes, decades of it

Long foundations.

✗ B. Only the last one: all did

✗ C. Only the first one: all did

✗ D. No, each one appeared without any connection to anything that researchers had done before: built on earlier

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

S150-F2-Q4 · question page 3149

AI.HISTORY.03 · See what each milestone changed and what it raised

Which new questions did these milestones bring?

Answer: C. Accuracy, cost and fairness

New questions.

- ✗ A. Only the color of computers: no
- ✗ B. Only typing speed: no
- ✗ D. None at all, because each milestone solved every problem and raised no new questions: they raised some

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

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S150-F2-Q5 · question page 3150

AI.HISTORY.03 · See what each milestone changed and what it raised

Which statement fits this course's approach?

Answer: D. Milestones are dated and each changed something specific

Dated, specific.

- ✗ A. History does not matter: no
- ✗ B. Dates are unknowable: they are known
- ✗ C. Every important change in AI happened in a single year: not accurate

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: hist_alexnet; tech_transformer; hist_chatgpt

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S151-M-Q1 · question page 3152

AI.HISTORY.04 · Tell narrow task performance from general intelligence claims

What is a narrow system?

Answer: C. One that does one task or a small set very well

Task-specific.

- ✗ A. A slow system: no
- ✗ B. A system that can do every task that any human can do, at least as well as any human: that is AGI
- ✗ D. A small computer: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S151-M-Q2 · question page 3153

AI.HISTORY.04 · Tell narrow task performance from general intelligence claims

Which is an example of a narrow system?

Answer: D. Deep Blue

Chess only.

- ✗ A. A human chess player: not a system
- ✗ B. The internet: no
- ✗ C. A system that can learn any new task as flexibly as people can, without special training: general

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S151-M-Q3 · question page 3154

AI.HISTORY.04 · Tell narrow task performance from general intelligence claims

What would general intelligence (AGI) mean?

Answer: B. Flexible ability across many tasks, including new ones

Broad flexibility.

- ✗ A. Being expensive: no
- ✗ C. Being fast: no
- ✗ D. Being extremely good at exactly one game, such as chess: narrow

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S151-M-Q4 · question page 3155

AI.HISTORY.04 · Tell narrow task performance from general intelligence claims

Does strong performance on one task show general intelligence?

Answer: A. No

Narrow skill.

- ✗ B. Yes, if the task is hard: no
- ✗ C. Yes, any system that is excellent at one task must be excellent at every other task: no
- ✗ D. Only for games: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S151-M-Q5 · question page 3156

AI.HISTORY.04 · Tell narrow task performance from general intelligence claims

Are definitions of AGI agreed?

Answer: C. No; they vary and are debated

Contested concept.

- ✗ A. Yes, there is one official definition of AGI that every researcher and company follows: debated
- ✗ B. There are none: some exist
- ✗ D. Yes, by law: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S151-B1-Q1 · question page 3157

AI.HISTORY.04 · Compare narrow and general ability with a calculator and a student

The calculator is like which kind of system?

Answer: D. Narrow

One task, superhuman.

- ✗ A. Neither: narrow
- ✗ B. Both: narrow
- ✗ C. General: narrow

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S151-B1-Q2 · question page 3158

AI.HISTORY.04 · Compare narrow and general ability with a calculator and a student

What can the calculator not do?

Answer: A. Learn a new game

No flexibility.

- ✗ B. Multiply large numbers quickly, which is the one thing that calculators struggle with: it can
- ✗ C. Turn on: it can
- ✗ D. Show numbers: it can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S151-B1-Q3 · question page 3159

AI.HISTORY.04 · Compare narrow and general ability with a calculator and a student

The student's strength is what?

Answer: B. Flexibility across many tasks

General ability.

- ✗ A. Never learning: no

✗ C. Never making mistakes: no

✗ D. Multiplying huge numbers faster: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S151-B1-Q4 · question page 3160

AI.HISTORY.04 · Compare narrow and general ability with a calculator and a student

Is being superhuman at one task the same as being general?

Answer: C. No

Different things.

✗ A. Yes, for calculators: no

✗ B. Only at night: no

✗ D. Yes, any system that beats people at one task is general by definition: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S151-B1-Q5 · question page 3161

AI.HISTORY.04 · Compare narrow and general ability with a calculator and a student

What is the lesson?

Answer: A. Narrow excellence is not general intelligence

Distinguish the two.

✗ B. Nothing: there is a lesson

✗ C. Students are calculators: no

✗ D. Calculators are smarter than students at everything: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S151-F1-Q1 · question page 3162

AI.HISTORY.04 · Evaluate narrow and general claims in practice

A headline says "AI beats doctors at reading one type of scan, so AI is now generally intelligent." What is wrong?

Answer: B. One task does not show general intelligence

Narrow result.

- ✗ A. Doctors cannot read scans: they can
- ✗ C. Nothing, because beating doctors at one task proves the system can do any task at all: narrow
- ✗ D. Scans are not tasks: they are

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S151-F1-Q2 · question page 3163

AI.HISTORY.04 · Evaluate narrow and general claims in practice

A chat model handles many tasks but fails at simple new puzzles. What does this suggest?

Answer: D. Its abilities are broad but uneven

Blurry line.

- ✗ A. That it cannot write: it can
- ✗ B. That puzzles are illegal: no
- ✗ C. That it must be completely general: uneven

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S151-F1-Q3 · question page 3164

AI.HISTORY.04 · Evaluate narrow and general claims in practice

Which question best tests a general-intelligence claim?

Answer: B. "Were the tasks new to the system?"

Novelty matters.

✗ A. "Is the logo new?": irrelevant

✗ C. "Is it expensive?": irrelevant

✗ D. "How many people liked the announcement?": irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S151-F1-Q4 · question page 3165

AI.HISTORY.04 · Evaluate narrow and general claims in practice

Is AlphaGo an example of general intelligence?

Answer: C. No; it was built for Go

Narrow.

✗ A. Yes, it plays all games: no

✗ B. It is a chatbot: no

✗ D. Yes: narrow

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S151-F1-Q5 · question page 3166

AI.HISTORY.04 · Evaluate narrow and general claims in practice

Which statement fits this course's approach?

Answer: A. Experts disagree about how close current systems are to AGI

Open debate.

✗ B. AGI is impossible, proven: not settled

✗ C. Experts all agree that full AGI was achieved and announced to the public in early 2026: not settled

✗ D. AGI exists already, proven: not settled

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S151-F2-Q1 · question page 3167

AI.HISTORY.04 · Read general-intelligence claims carefully

A company claims its product "reached AGI." What should you ask first?

Answer: D. "How is general defined here?"

Definition first.

- ✗ A. "Can I buy it today at the lowest possible price, before everyone else hears about it?": not the key
- ✗ B. "What color is it?": irrelevant
- ✗ C. "Is it fast?": not the key

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S151-F2-Q2 · question page 3168

AI.HISTORY.04 · Read general-intelligence claims carefully

Why ask whether the claimant benefits?

Answer: B. Incentives can shape claims

Consider incentives.

- ✗ A. Because companies always tell the exact truth about their own products in every announcement: not always
- ✗ C. Because it is polite: not the reason
- ✗ D. Because benefits are illegal: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S151-F2-Q3 · question page 3169

AI.HISTORY.04 · Read general-intelligence claims carefully

Why ask how a system fails?

Answer: D. Failures reveal limits

Limits matter.

- ✗ A. Because failures are rare: not the reason
- ✗ B. Because it is fun: no
- ✗ C. Because systems that fail are always narrow, and systems that never fail are always general: too simple

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S151-F2-Q4 · question page 3170

AI.HISTORY.04 · Read general-intelligence claims carefully

Do experts agree that AGI is well defined?

Answer: C. No

Contested.

- ✗ A. Yes, by law: no
- ✗ B. Only in movies: no
- ✗ D. Yes: debated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S151-F2-Q5 · question page 3171

AI.HISTORY.04 · Read general-intelligence claims carefully

Does this course settle the AGI debate?

Answer: A. No

Open question.

- ✗ B. Yes, it says AGI exists: no verdict
- ✗ C. Yes, it says AGI is impossible: no verdict
- ✗ D. Yes, it states the exact year: no verdict

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S152-M-Q1 · question page 3173

AI.HISTORY.05 · Tell recorded events from predictions and judge capability claims

In this fictional headline, which part is a recorded event?

Prepared example (a fictional headline)

Headline: "New AI released Tuesday aces math olympiad problems; experts say it will make math teachers obsolete by 2030."

Answer: D. Released Tuesday

A checkable event.

- ✗ A. Aces math olympiad problems: capability claim
- ✗ B. None: one is
- ✗ C. Will make math teachers obsolete: prediction

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S152-M-Q2 · question page 3174

AI.HISTORY.05 · Tell recorded events from predictions and judge capability claims

In the same headline, which part is a capability claim?

Prepared example (a fictional headline)

Headline: "New AI released Tuesday aces math olympiad problems; experts say it will make math teachers obsolete by 2030."

Answer: C. Aces math olympiad problems

Needs evidence.

✗ A. None: one is

✗ B. Released Tuesday, which describes what the new AI system is able to do for its users: event

✗ D. Obsolete by 2030: prediction

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S152-M-Q3 · question page 3175

AI.HISTORY.05 · Tell recorded events from predictions and judge capability claims

In the same headline, which part is a prediction?

Prepared example (a fictional headline)

Headline: "New AI released Tuesday aces math olympiad problems; experts say it will make math teachers obsolete by 2030."

Answer: B. Teachers obsolete by 2030

About the future.

✗ A. Released Tuesday, since it describes what will happen to the system in the near future: event

✗ C. Aces math problems: capability

✗ D. None: one is

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S152-M-Q4 · question page 3176

AI.HISTORY.05 · Tell recorded events from predictions and judge capability claims

What do capability claims need?

Answer: A. Evidence: which tests, how many, who ran them

Test details.

✗ B. A famous logo: no

✗ C. A long article: no

✗ D. A confident tone and an exciting headline: not evidence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S152-M-Q5 · question page 3177

AI.HISTORY.05 · Tell recorded events from predictions and judge capability claims

Does a confident tone turn a prediction into a fact?

Answer: C. No

Predictions stay uncertain.

✗ A. Only in headlines: no

✗ B. Yes, if repeated: no

✗ D. Yes: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S152-B1-Q1 · question page 3178

AI.HISTORY.05 · Sort headline statements into three kinds

"Launched last week" is which kind?

Answer: D. Recorded event

Happened.

✗ A. Value: no

✗ B. Capability claim: no

✗ C. Prediction: past event

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S152-B1-Q2 · question page 3179

AI.HISTORY.05 · Sort headline statements into three kinds

"Detects skin cancer as well as doctors" is which kind?

Answer: A. Capability claim

Needs tests.

- ✗ B. Prediction: present ability
- ✗ C. Value: no
- ✗ D. Recorded event: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S152-B1-Q3 · question page 3180

AI.HISTORY.05 · Sort headline statements into three kinds

"Will save millions of lives" is which kind?

Answer: B. Prediction

Future.

- ✗ A. Recorded event: not happened
- ✗ C. Value: no
- ✗ D. Capability claim: future

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S152-B1-Q4 · question page 3181

AI.HISTORY.05 · Sort headline statements into three kinds

What checks a recorded event?

Answer: A. Records

Documentation.

- ✗ B. Tone: no

✗ C. Likes: no

✗ D. Expert predictions about the future, which confirm whether an event really happened: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S152-B1-Q5 · question page 3182

AI.HISTORY.05 · Sort headline statements into three kinds

What checks a capability claim?

Answer: D. Test evidence

Tests.

✗ A. The number of times the claim has been repeated in different news articles online: not evidence

✗ B. Predictions: no

✗ C. Records of the launch date: not capability

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S152-F1-Q1 · question page 3183

AI.HISTORY.05 · Evaluate AI headlines in practice

A company says its model "matches experts," based on its own unpublished test. How strong is this evidence?

Answer: B. Weak until details or independent checks appear

Needs details.

✗ A. Irrelevant: it is a claim

✗ C. Very strong, because companies always test their own products in the most careful possible way: unverified

✗ D. Proof: not proof

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: none (stable)

S152-F1-Q2 · question page 3184

AI.HISTORY.05 · Evaluate AI headlines in practice

A prediction says "AI will do X by 2030." What should you check?

Answer: C. Who predicts, their track record and assumptions

Source and assumptions.

- ✗ A. The font: irrelevant
- ✗ B. Nothing: check
- ✗ D. Whether the prediction has been shared widely: not evidence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S152-F1-Q3 · question page 3185

AI.HISTORY.05 · Evaluate AI headlines in practice

A demo video shows a robot folding laundry perfectly. What should you consider?

Answer: B. It may show the best attempt out of many

Demos can be selected.

- ✗ A. Nothing: consider selection
- ✗ C. That robots cannot fold: not shown
- ✗ D. That the robot folds every piece of laundry perfectly every single time in any home: selective

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S152-F1-Q4 · question page 3186

AI.HISTORY.05 · Evaluate AI headlines in practice

Other researchers repeat a test and get the same result. What does this do?

Answer: D. Strengthens the claim

Independent replication.

- ✗ A. Nothing: it helps
- ✗ B. Proves it forever: too strong
- ✗ C. Weakens the claim: strengthens

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S152-F1-Q5 · question page 3187

AI.HISTORY.05 · Evaluate AI headlines in practice

What is the first step with a surprising AI headline?

Answer: C. Find the original source

Go to the source.

- ✗ A. Ignore all news: too strong
- ✗ B. Believe it: check first
- ✗ D. Share it with friends quickly: check first

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S152-F2-Q1 · question page 3188

AI.HISTORY.05 · Use history for perspective on predictions

Have AI predictions historically been wrong?

Answer: A. Yes, in both directions

Uncertain forecasting.

- ✗ B. Only too early: both directions
- ✗ C. Only too late: both directions
- ✗ D. No, every prediction about AI has come true exactly on the date that was predicted: often wrong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S152-F2-Q2 · question page 3189

AI.HISTORY.05 · Use history for perspective on predictions

What might a press release leave out?

Answer: B. Failures

Selective reporting.

- ✗ A. The company name: no
- ✗ C. The date: no
- ✗ D. Every success: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S152-F2-Q3 · question page 3190

AI.HISTORY.05 · Use history for perspective on predictions

How should trust be assigned to statements in a headline?

Answer: D. Each according to its evidence

Evidence-based trust.

- ✗ A. None at all: too strong
- ✗ B. Fully to predictions: no
- ✗ C. Equally to every statement: not equal

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S152-F2-Q4 · question page 3191

AI.HISTORY.05 · Use history for perspective on predictions

Which strengthens a capability claim most?

Answer: C. Independent researchers repeating the test

Replication.

- ✗ A. The company's chief executive repeating the claim confidently in several interviews: not independent
- ✗ B. More advertising: no
- ✗ D. A larger font: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S152-F2-Q5 · question page 3192

AI.HISTORY.05 · Use history for perspective on predictions

Which lesson does AI history support about predictions?

Answer: A. Read them with caution

Caution.

- ✗ B. Believe the most confident ones, since confidence is the best sign of an accurate prediction: not supported
- ✗ C. Never read them: too strong
- ✗ D. They are always right: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S153-M-Q1 · question page 3194

AI.HISTORY.06 · Describe AI energy and resource use with dated, uncertain estimates

What is a data centre?

Answer: A. A building full of computers

Computing facility.

- ✗ B. A school computer lab: too small
- ✗ C. A website: no

✗ D. A shopping center where people buy computers and phones for their homes and offices: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: energy_iea

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S153-M-Q2 · question page 3195

AI.HISTORY.06 · Describe AI energy and resource use with dated, uncertain estimates

Per the IEA, about how much electricity did data centres use in 2024?

Answer: D. About 415 TWh

Dated estimate.

✗ A. Exactly 0 TWh: no
✗ B. About 1.5 TWh: that is a percent
✗ C. About 945 TWh: that is 2030 projection

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: energy_iea

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S153-M-Q3 · question page 3196

AI.HISTORY.06 · Describe AI energy and resource use with dated, uncertain estimates

About what share of global electricity was that?

Answer: C. About 1.5 percent

IEA estimate.

✗ A. About 15 percent: no
✗ B. About 0.01 percent: too low
✗ D. About 50 percent: far too high

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: energy_iea

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S153-M-Q4 · question page 3197

AI.HISTORY.06 · Describe AI energy and resource use with dated, uncertain estimates

From 415 to 945 TWh is roughly what increase?

Answer: B. About 530 TWh

945 - 415 = 530.

- ✗ A. About 2.3 TWh: ratio-ish
- ✗ C. About 945 TWh: that is the endpoint
- ✗ D. About 1,360 TWh: added

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: energy_iea

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S153-M-Q5 · question page 3198

AI.HISTORY.06 · Describe AI energy and resource use with dated, uncertain estimates

Why not quote one universal per-question energy number?

Answer: D. Figures vary widely

Too variable.

- ✗ A. Because AI uses no energy: it does
- ✗ B. Because energy is not measured: it is
- ✗ C. Because every AI question uses exactly the same energy, so no number is needed at all: they vary

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: energy_iea

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S153-B1-Q1 · question page 3199

AI.HISTORY.06 · Compare AI energy estimates with estimating a car trip's fuel

What does a trip's fuel use depend on?

Answer: A. Car, distance, speed, traffic and weather

Many factors.

- ✗ B. Only the driver's name: no

✗ C. Nothing: many things

✗ D. Only the color of the car, which always decides how much fuel any trip will use: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: energy_iea

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S153-B1-Q2 · question page 3200

AI.HISTORY.06 · Compare AI energy estimates with estimating a car trip's fuel

Is "a trip uses 5 liters" meaningful alone?

Answer: B. No; details are missing

Needs context.

✗ A. Yes, for short trips: no

✗ C. Only in summer: no

✗ D. Yes: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: energy_iea

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S153-B1-Q3 · question page 3201

AI.HISTORY.06 · Compare AI energy estimates with estimating a car trip's fuel

In AI, what is like the car?

Answer: C. The model and hardware

What does the work.

✗ A. The person asking the question, whose typing speed decides the energy that is used: no

✗ B. The question's topic: not mainly

✗ D. The weather: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: energy_iea

S153-B1-Q4 · question page 3202

AI.HISTORY.06 · Compare AI energy estimates with estimating a car trip's fuel

What should good estimates state?

Answer: D. Assumptions and ranges

Transparency.

- ✗ A. Only the source's logo: no
- ✗ B. Only the date: not enough
- ✗ C. Only one exact number: not good

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: energy_iea

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S153-B1-Q5 · question page 3203

AI.HISTORY.06 · Compare AI energy estimates with estimating a car trip's fuel

What is the lesson?

Answer: A. Energy figures depend on many factors

Context matters.

- ✗ B. Cars use no fuel: no
- ✗ C. AI uses no energy: no
- ✗ D. AI energy use can be described by one exact universal number for every question: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: energy_iea

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S153-F1-Q1 · question page 3204

AI.HISTORY.06 · Read energy claims in practice

A post says "Every AI question uses exactly one bottle of water." What should you check?

Answer: B. Its source, date and assumptions

Particular assumptions.

- ✗ A. The bottle's brand: irrelevant
- ✗ C. Nothing, since a claim that is easy to remember must be accurate for every AI question: check
- ✗ D. The post's likes: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: energy_iea

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S153-F1-Q2 · question page 3205

AI.HISTORY.06 · Read energy claims in practice

Do data centres serve only AI?

Answer: C. No; they serve much more

Many services.

- ✗ A. Yes, every data centre in the world was built only to run AI chatbots and nothing else: more uses
- ✗ B. Only in Europe: no
- ✗ D. Yes, since 2022: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: energy_iea

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S153-F1-Q3 · question page 3206

AI.HISTORY.06 · Read energy claims in practice

The IEA's 2030 figure is what kind of statement?

Answer: B. A projection with large uncertainty

Base-case projection.

- ✗ A. A law: no
- ✗ C. A recorded fact about electricity use that has already been measured in the year 2030: future

✗ D. A guess with no method: it has a method

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: energy_iea

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S153-F1-Q4 · question page 3207

AI.HISTORY.06 · Read energy claims in practice

A student writes "AI uses 1.5% of world electricity." What is the correction?

Answer: A. The IEA estimate is for all data centres, not just AI

Scope matters.

✗ B. AI uses none: no

✗ C. No correction is needed, since every data centre is used only for AI and nothing else: more than AI

✗ D. It is 50%: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: energy_iea

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S153-F1-Q5 · question page 3208

AI.HISTORY.06 · Read energy claims in practice

Which resource besides electricity can AI use?

Answer: C. Water for cooling in some designs

Cooling water.

✗ A. Gasoline, which every data centre burns directly to power the computers that run AI: not typical

✗ B. Sand only: no

✗ D. Paper: not significant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: energy_iea

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S153-F2-Q1 · question page 3209

AI.HISTORY.06 · Understand why energy estimates differ

Which factor affects AI energy estimates?

Answer: D. The electricity source

Assumptions matter.

- ✗ A. The user's age: no
- ✗ B. The day of the week only: not mainly
- ✗ C. The number of letters: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: energy_iea

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S153-F2-Q2 · question page 3210

AI.HISTORY.06 · Understand why energy estimates differ

How can newer hardware affect energy?

Answer: B. It can use less energy per task

Efficiency.

- ✗ A. It has no effect: it can
- ✗ C. It uses no energy: no
- ✗ D. It always uses far more energy per task, since newer computers are always less efficient: often less

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: energy_iea

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S153-F2-Q3 · question page 3211

AI.HISTORY.06 · Understand why energy estimates differ

Why might total energy still rise with more efficient hardware?

Answer: C. Growing use can raise the total

More use.

✗ A. It cannot rise: it can

✗ B. Because efficient hardware always uses more energy in total than older, less efficient hardware: not the reason

✗ D. Because energy is free: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: energy_iea

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S153-F2-Q4 · question page 3212

AI.HISTORY.06 · Understand why energy estimates differ

Training versus everyday use: why does this matter for estimates?

Answer: D. They count different things

What is counted.

✗ A. Because training is free: no

✗ B. It does not matter: it does

✗ C. Because training and everyday use always consume exactly the same amount of energy: they differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: energy_iea

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S153-F2-Q5 · question page 3213

AI.HISTORY.06 · Understand why energy estimates differ

What should you look for in an energy claim?

Answer: A. Source, date, what was measured and uncertainty

Full context.

✗ B. Only a single number: not enough

✗ C. Only the headline: not enough

✗ D. Only how alarming or reassuring the claim sounds: not evidence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: energy_iea